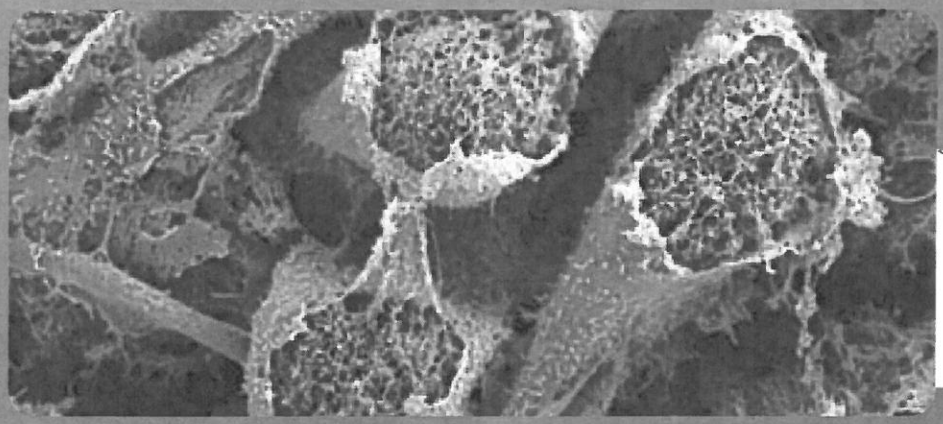


Figure 2.4.14

Stem cells seen under a scanning electron microscope

Stem cells extracted from the hair root are grown in culture for about two weeks. They are completely surrounded by liquid and divide, forming small masses of cells. After two weeks the upper surface of the cell mass is exposed to the air. This is the trigger for the stem cells to turn into skin cells and skin like that in Figure 2.4.15.

The skin is then grafted onto the wound. Within a few days, doctors will know if the graft is going to work. A successful skin graft will look just like your old skin within 10 weeks.

In traditional skin grafts, healthy skin is taken from another part of your body to put over a wound. You then end up with two scars. Which way would you prefer—the traditional method of cell culture or the latest technology?

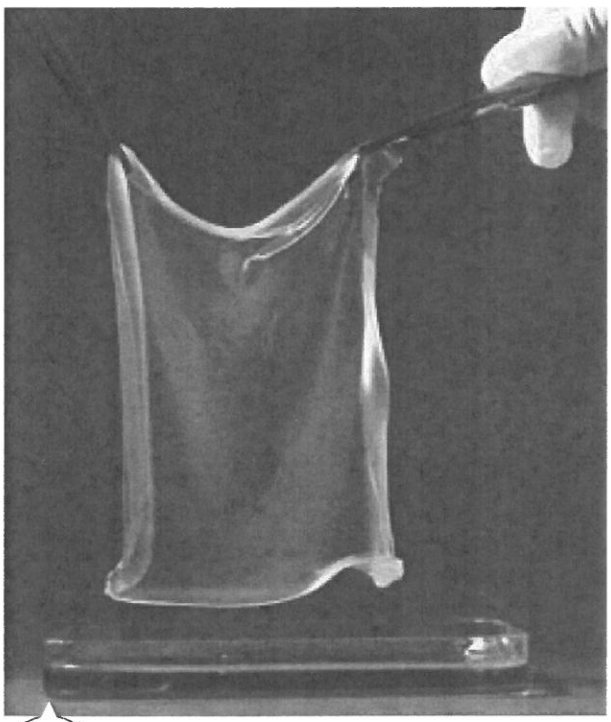


Figure 2.4.15 A strip of skin is held above the culture medium.

Growing organs

Scientists hope that in the future complete organs can be grown for transplant. This is becoming closer to reality.

There are children in the USA who have new bladders that were grown from their own cells. These children were born with defective bladders. Doctors took two types of cells from the child's own bladder—cells from the lining of the bladder, and muscle cells from the outside. These cells were grown in culture solution and when there were enough cells they were placed on a bladder-shaped shell made from a biodegradable material. The lining cells went on the inside of the shell and the muscle cells on the outside. The new bladder continued to grow in culture for another seven weeks. After that it was sutured (stitched) into the body of the patients where the new bladders continued to grow and remain healthy.

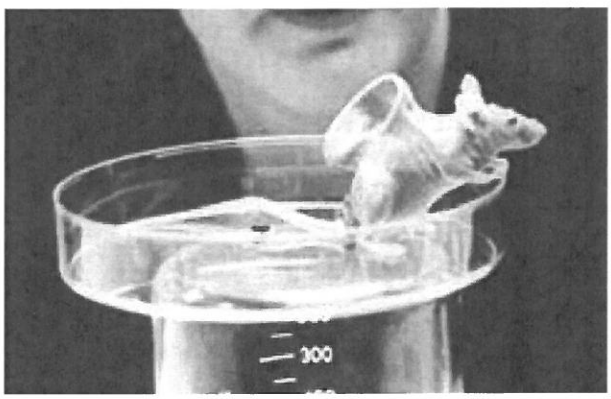


Figure 2.4.16 The ear on the back of this mouse was grown from cartilage cells from a cow implanted on a mould in the shape of an ear. The experiment suggests that replacement body parts may be able to be grown in the future.

2.8

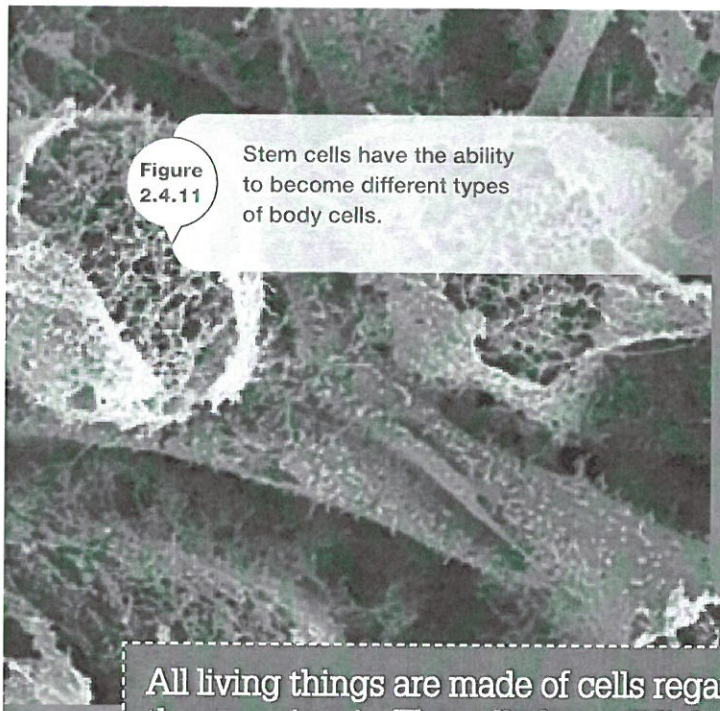


Figure 2.4.11 Stem cells have the ability to become different types of body cells.

SCIENCE AS A HUMAN ENDEAVOUR

Use and influence of science

Growing cells

All living things are made of cells regardless of how big or small, simple or complex the organism is. The cells from all living things have many features in common and so the study of cells can provide information about all living things.

In the last 100 years scientists have learnt how to grow cells separate from the organism they came from. This process is called **cell culture**. The cells are kept at the correct temperature and in a solution that provides them with all the minerals and nutrients they require to keep them alive. You can see a scientist working with cultured cells in Figure 2.4.12.

Cultured cells have been used to produce the vaccines that protect us from diseases such as measles, rubella, mumps, chickenpox and polio. These cells are also used to produce agents that can help fight cancer.

Most cultured cells are not from humans. However, there are times when human cells are cultured. The bottles in which they are cultured are shown in Figure 2.4.13.

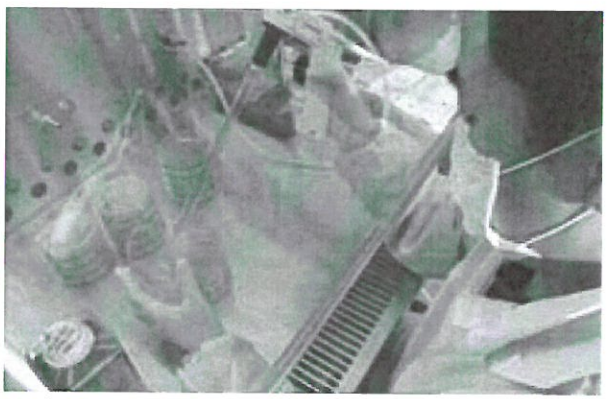


Figure 2.4.12 This researcher is working in a tissue culture laboratory.



Figure 2.4.13 Tissues are cultured in flat bottles like these.

Growing tissue

When your body is damaged or new cells are required to replace ones that are worn out, the cells in your body divide. Skin cells divide to produce new skin cells, and muscle cells produce new muscle cells. In the human body, there are also special cells called **stem cells**. Stem cells are able to become different types of cells under special conditions. They can be found in the roots of your hair. Cells from a few hairs could be very useful if you had a large wound such as a burn. You can see some stems cells in Figure 2.4.14 on page 70.